



The Long-term Photometric Behavior of 39 Semiregular Variable Stars

Robert R. Cadmus, Jr.

Department of Physics, Grinnell College 1115 8th Avenue Grinnell, IA 50112, USA; cadmus@grinnell.edu

Received 2023 September 19; revised 2024 January 30; accepted 2024 January 31; published 2024 April 8

Abstract

Photometric measurements of the light and color variations of 39 semiregular variable stars over a 30 yr time interval have been used to explore the systematics of these variations. These results show the complex nature of the frequency compositions of the light curves of these stars. The frequencies present in the light curves tend to be harmonic in nature, suggesting that both modes of pulsation and shape effects may be involved and the nature of the variations indicates that stochastic excitation is involved.

Unified Astronomy Thesaurus concepts: Long period variable stars (935); Semi-regular variable stars (1444); Stellar pulsations (1625); Pulsation modes (1309); Multi-periodic pulsation (1078); Photometry (1234)

Supporting material: data behind figure

1. Introduction

Semiregular variable stars are interesting for a number of reasons. They are pulsating, have complex atmospheric motions, and are cool enough for complicated chemistry to occur. They also display complex light-curve variations on multiple timescales in addition to an underlying regular periodicity. These stars are red giants that are in the asymptotic giant branch phase of their lives during which they become unstable, leading to luminosity and spectral variations. The prevailing view is that these stars are pulsating and that the details of the light curves can be used to learn about that process. The goal of the Grinnell photometric program was to obtain high-quality light curves in order to better characterize that behavior. In particular, we have investigated the role that stochastic excitation and damping play in the oscillations of these stars.

Most of the stars described here were selected based on prior occurrence of conspicuous episodes of reduced light-curve amplitude. Subsequent developments in the understanding of these stars show that these episodes may simply be random fluctuations with no further significance but the existence of a selection criterion should be kept in mind. The work presented here involves dozens of stars but the emphasis is on what can be learned from them collectively rather than individually. Our investigation of RS Cyg has already been published (Cadmus 2022) and work on other stars is continuing.

2. Background

The process of trying to unravel the mysteries of semiregular variable stars has a long history and a great deal of observational and theoretical work has been done. The idea that the luminosity variations originate in the pulsation of the stellar atmosphere has been clarified by insights from astroseismology. The discovery of acoustically driven oscillations in the Sun naturally led to the search for evidence of stochastic excitation in other stars, including the semiregular variables (Kjeldsen & Bedding 1995; Christensen-Dalsgaard

et al. 2001; Bedding & Kjeldsen 2003). Evidence for such behavior has been detected in several ways. Bedding (2003), Bedding et al. (2005), Kiss et al. (2006), and Cunha et al. (2020) have used the fact that stochastic driving, presumably as a result of convection in the outer layers of these stars, spreads a peak in the Fourier spectrum of a star's light curve into a family of peaks whose distribution of strengths can be represented by a Lorentzian function with a width that is related to the damping time and can be used to calculate the lifetime of the mode. Stochastic behavior can also be revealed by variations in the amplitude of the light curve (Kjeldsen & Bedding 1995; Christensen-Dalsgaard et al. 2001) and by the variation of the observed light-curve amplitude with the length of the run of photometric data considered (Yu et al. 2020). Stochastic processes are believed to play a more important role in the types of stars with less regular behavior, including the semiregulars, and a less significant role in the more regular stars, such as the Miras (Dziembowski et al. 2001; Mosser et al. 2013; Yu et al. 2020), and to become less important as the pulsational period increases (Bányai et al. 2013; Cunha et al. 2020; Yu et al. 2020).

Semiregular stars are apparently capable of pulsating in different modes, which may coexist, and the frequency composition of the light curves has been explored extensively using Fourier analysis and other techniques (Mattei & Foster 1997; Percy & Polano 1998; Kiss et al. 1999; Percy et al. 2003; Percy & Tan 2013). A period ratio of 2:1, or slightly smaller, has often been reported (Mattei & Foster 1997; Kiss et al. 1999).

3. Observations

Thirty-nine stars were observed between 1984 and 2022 (JD 2,445,849–JD 2,459,608) and are listed in Table 1. The selection process yielded a few stars that are not classified as semiregular and a number that did not exhibit any dramatic episodes of reduced amplitude during the duration of this project. The data for these stars, as well as those for stars that were observed for only a limited period of time, are included here in order to make our observational results for them available. Table 1 also gives the Tycho catalog designations (Høg et al. 2000) and V and $B - V$ values for the comparison stars taken from SIMBAD, except in four cases for which



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](https://creativecommons.org/licenses/by/4.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Table 1
Comparison Stars, Frequencies, and Frequency Ratios

Star	Comparison	V	$B - V$	f_1 (day ⁻¹)	P_1 (days)	f_2/f_1	f_3/f_1
EU And	3636-2089-1	7.78	1.63	0.00377	266	1.68	...
RU And	2814-1513-1	9.93	1.47	0.00407	246	1.97	...
RV And	3285-42-1	8.39	1.11	0.00595	168	1.88	2.87
S Aql	1618-1558-1	9.69	0.46	0.00682	147	2.10	...
RS Aqr ^a	5201-0192-1	10.01	1.10	0.00451	222	1.96	...
S Aur	2411-1188-1	9.49	1.76	0.00172	582	2.00	2.99
RW Boo	2554-1351-1	7.95	0.62	0.00244	410	2.07	7.15
RX Boo	2011-1146-1	6.26	0.37	0.00620	161	1.62	...
U Boo	1481-0504-1	10.30	0.39	0.00492	203	2.01	...
V Boo	3036-209-1	9.16	0.88	0.00385	260	1.93	2.89
U Cam	4067-72-1	8.06	0.05	0.00254 ^b	394 ^b	1.81 ^b	...
WZ Cas	4014-1936-1	7.29	0.10	0.00272	368	1.95	...
RS Cnc	2495-846-1	6.00	0.56	0.00417	240	1.93 ^b	...
V CVn	3460-1403-1	7.68	1.13	0.00529 ^b	189 ^b	1.95 ^b	...
RS Cyg	3151-1358-1	7.10	0.51	0.00237	422	1.99	2.99
RU Cyg	3971-1874-1	6.96	1.29	0.00230	434	1.84 ^b	...
TT Cyg	2659-1807-1	7.86	0.06	0.00499 ^b	200 ^b	1.74 ^b	...
V778 Cyg	3962-1544-1	9.57	1.28	0.00331	302	1.90	2.93
W Cyg	3590-3349-1	5.22	0.97	0.00387	259	1.96	...
U Del	1638-1025-1	6.24	0.94	0.00833	120	1.53	...
RY Dra	4169-93-1	6.54	1.10	0.00364	275	1.78	2.70
UX Dra	4587-2406-1	6.57	0.38	0.00302	332	1.87	...
SX Her	2035-238-1	9.26	0.37	0.00966	104	1.98	...
X Her	3494-1307-1	6.64	-0.03	0.00572	175	1.72	...
RT Hya ^a	4861-109-1	7.36	0.95	0.00395	253	1.89	...
RS Lac	3211-0412-1	9.44	1.55	0.00419	238	2.01	...
X Lib ^a	6196-0048-1	10.65	0.62	0.00322	310	1.98	2.77
U LMi	2508-1039-1	10.04	0.44	0.00367	272	1.91	...
X Mon	5380-1720-1	7.12	1.50	0.00640	156	...	...
RV Peg ^a	2734-42-1	9.49	0.44	0.00258	387	2.02	...
S Per	3698-1830-1	8.32	1.02	0.00124	807	2.59	...
U Per	3689-1381-1	8.80	1.13	0.00314	319	2.00	3.00
RW Sgr ^a	6304-0290-1	9.98	0.11	0.00543	184	2.00	...
W Tau	1265-0016-1	10.39	0.92	0.00411 ^b	244 ^b	1.90 ^b	2.88 ^b
RZ UMa	4132-1977-1	8.54	1.15	0.00365	274	1.92 ^b	...
ST UMa	3015-1713-1	6.68	0.88	0.01160 ^c	86 ^c	note ^c	...
Z UMa	3839-613-1	8.44	0.65	0.00528	189	1.94	3.01
R UMi	4423-481-1	8.47	0.92	0.00310	323	1.90	...
SW Vir	4961-229-1	8.66	0.42	0.00646	155	2.00	...

Notes.^a Stars with limited data.^b Means of the two components of a doublet in the Fourier spectrum.^c See the text.

SIMBAD data were not available. In those cases (the comparison stars for RX Boo, RS Cnc, RY Dra, and ST UMa) values measured at Grinnell, which in other cases agree with those in SIMBAD to within a few hundredths of a magnitude, are shown. The observations relevant to this paper consisted of V - and B -band differential photometry acquired with the 0.61 m telescope at the Grant O. Gale Observatory of Grinnell College using an uncooled 1P21 photomultiplier. The data were corrected for differential extinction and transformed to the Johnson UB V system. In order to minimize the effects of annual breaks in the light curves, some data were taken in twilight and at high air mass, but a correction was made for the nonlinear variation of sky brightness during a single star sequence. The resulting ΔV (black) and $\Delta(B - V)$ (red) light curves are shown in Figures 1–10. For X Lib the observations were so limited that they are not included in the figures. In these figures the vertical scale factors are the same for the ΔV

and $\Delta(B - V)$ data for each star. The variability type and spectral type of each variable from the AAVSO International Variable Star Index database¹ are also included in these figures. The reliability of the data was addressed in Cadmus (2021). The ΔV and $\Delta(B - V)$ data in Figures 1–10 can be converted to V and $B - V$ values by adding them to the comparison star values in Table 1.

To facilitate subsequent analysis, the observational data were represented by a “guided” spline fit that resulted in a dense set of evenly spaced points with no annual breaks. This was done by generating a spline fit not to the data points themselves, but to a set of guide points that were adjusted by hand to steer the fit until it coincided with the data. As shown in Figures 1–10, the resulting curves accurately represent the significant features of the data. The few points that lie well off the curves are ones

¹ <http://www.aavso.org/vsx>

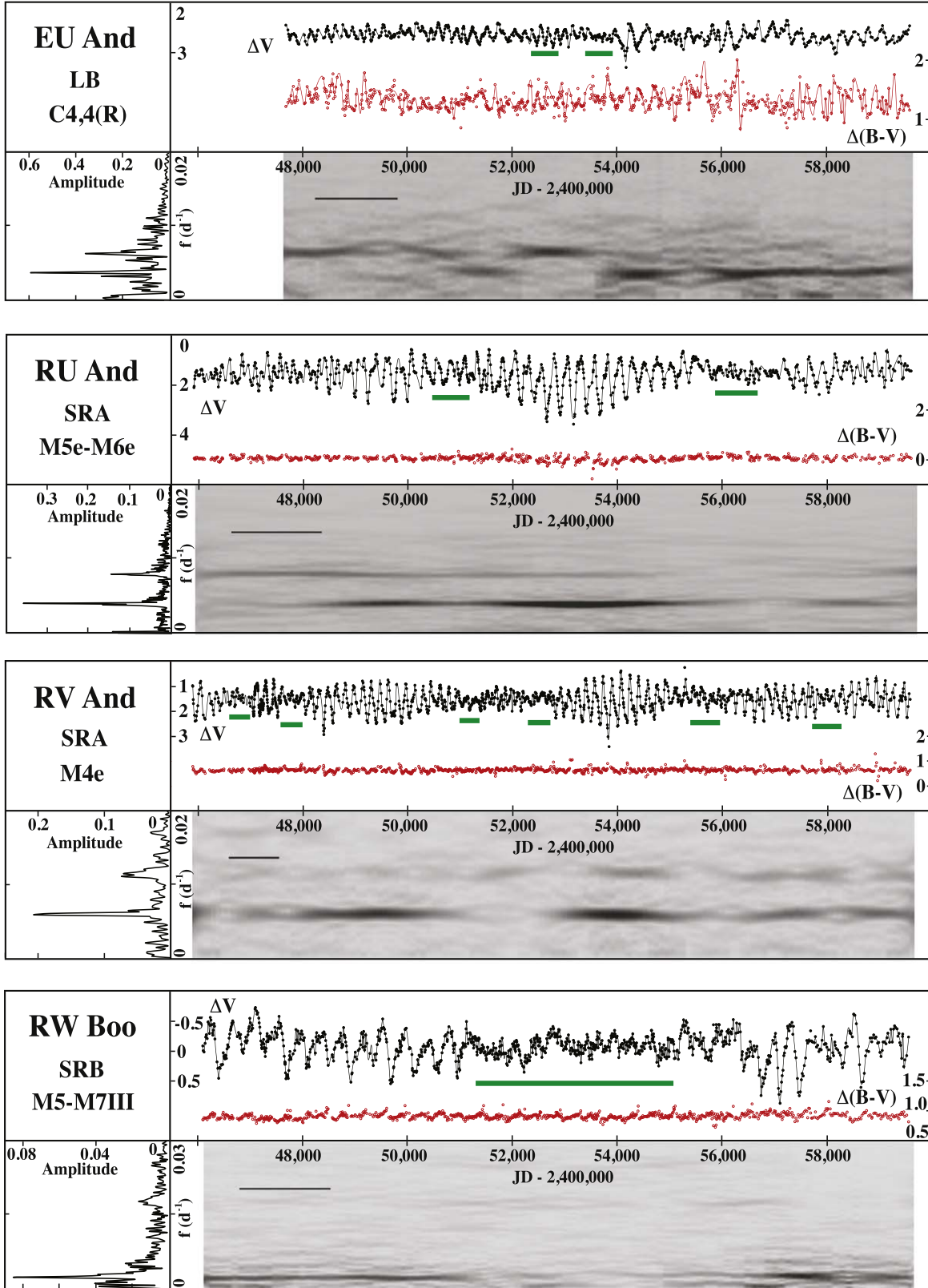


Figure 1. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that involve a frequency ($1/\text{period}$) change. Upper panel: the ΔV and $\Delta(B - V)$ light curves. The photometric measurements are shown with small symbols and the spline fit with the curve. The green bars indicate the approximate locations of the episodes of reduced amplitude that are discussed in the text. Lower left panel: the Fourier spectrum of the entire ΔV light curve. Lower right panel: the map of Fourier spectrum strength vs. frequency and Julian date. More explanation is provided in the text. The differential light curves for Figures 1–10 are available as Data Behind the Figure.

(The data used to create this figure are available.)

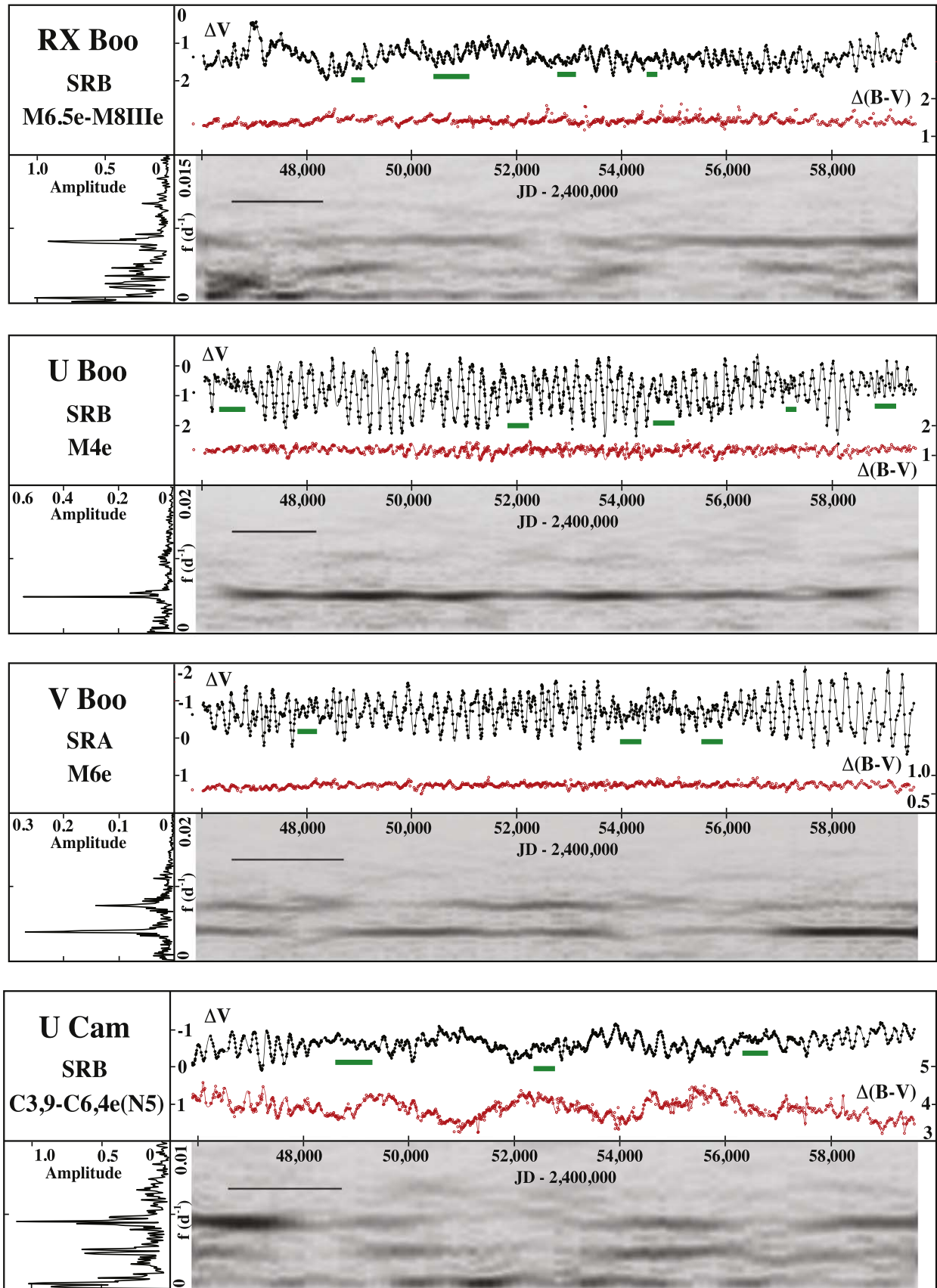


Figure 2. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

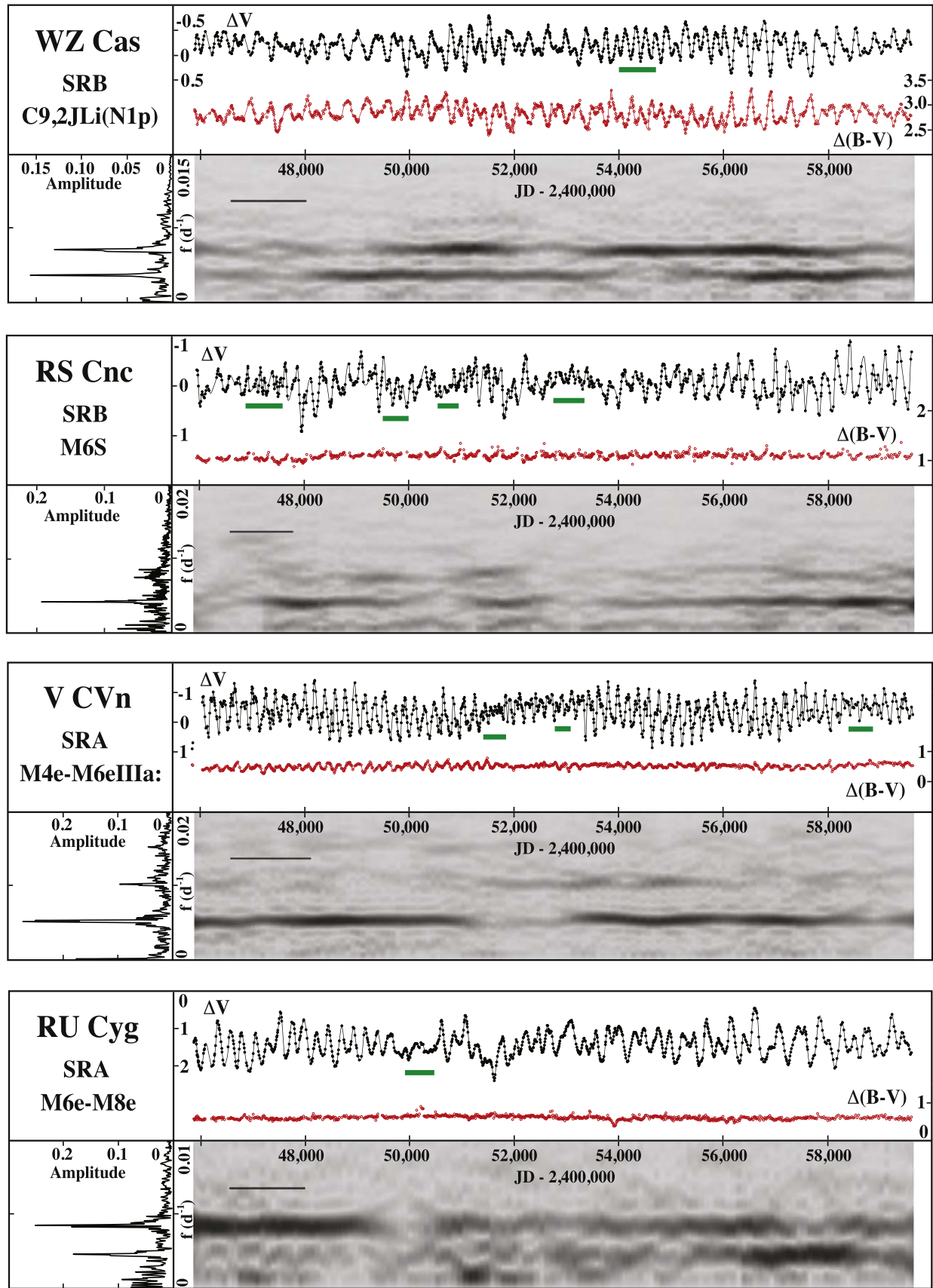


Figure 3. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

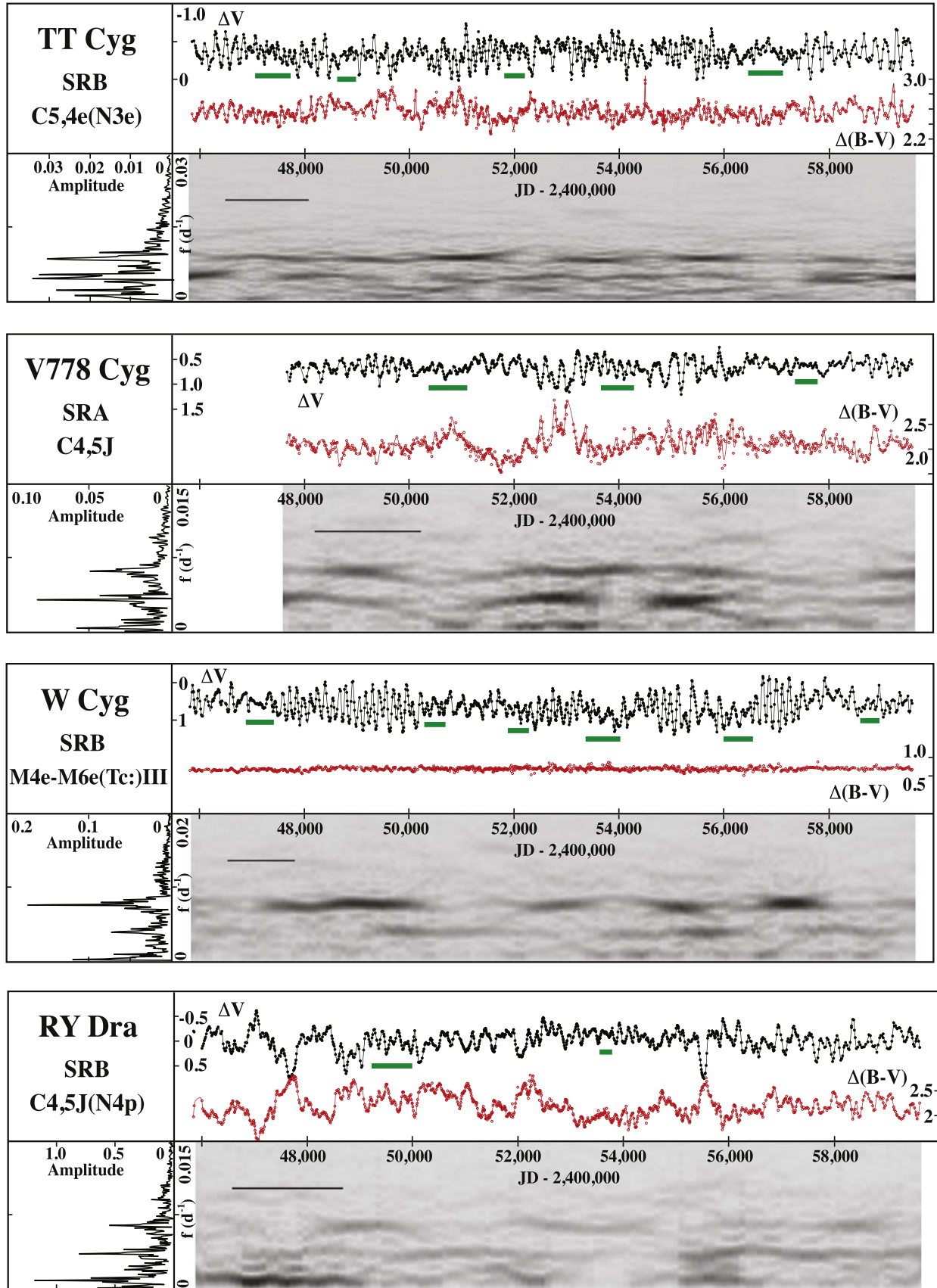


Figure 4. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

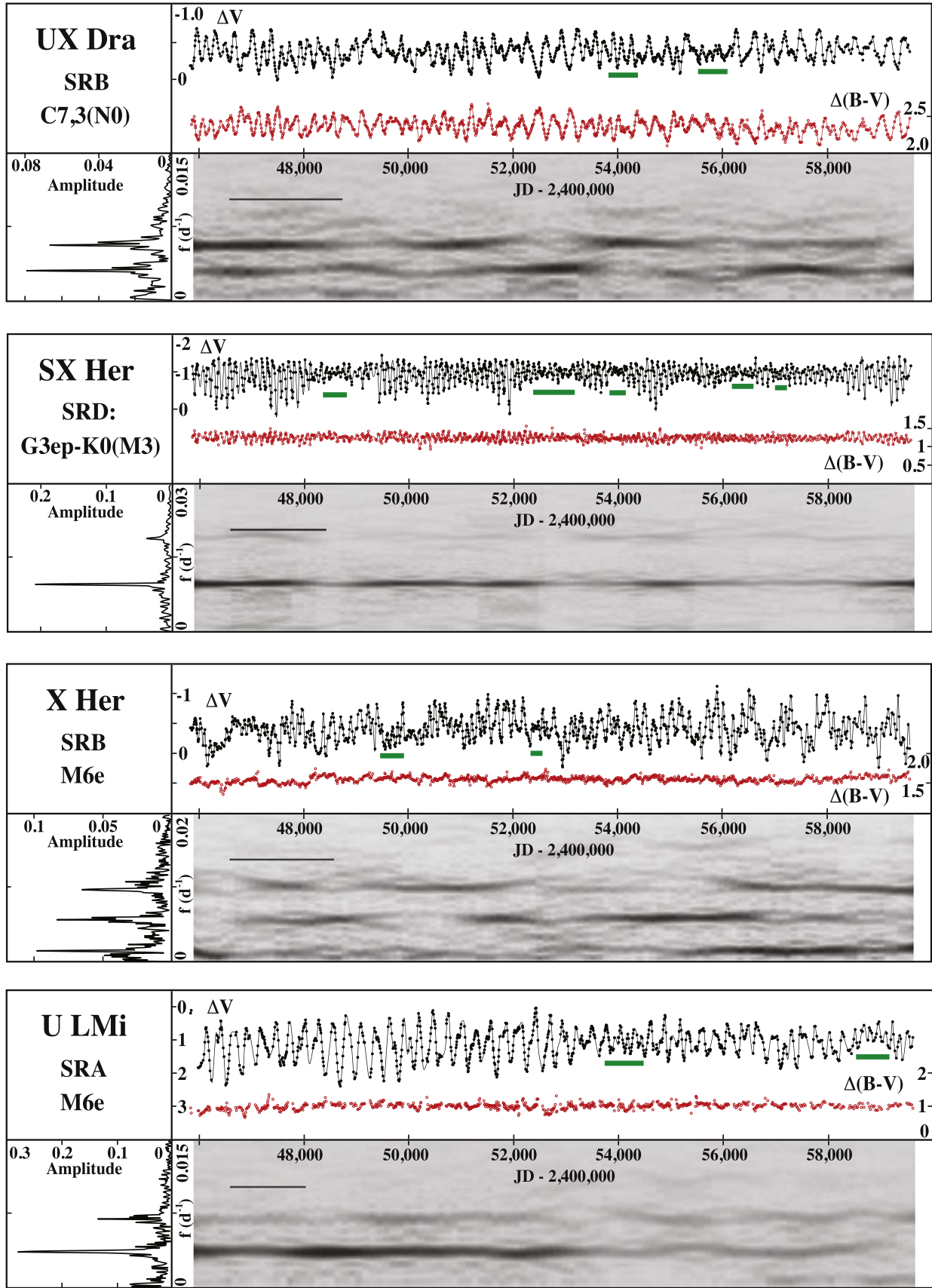


Figure 5. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

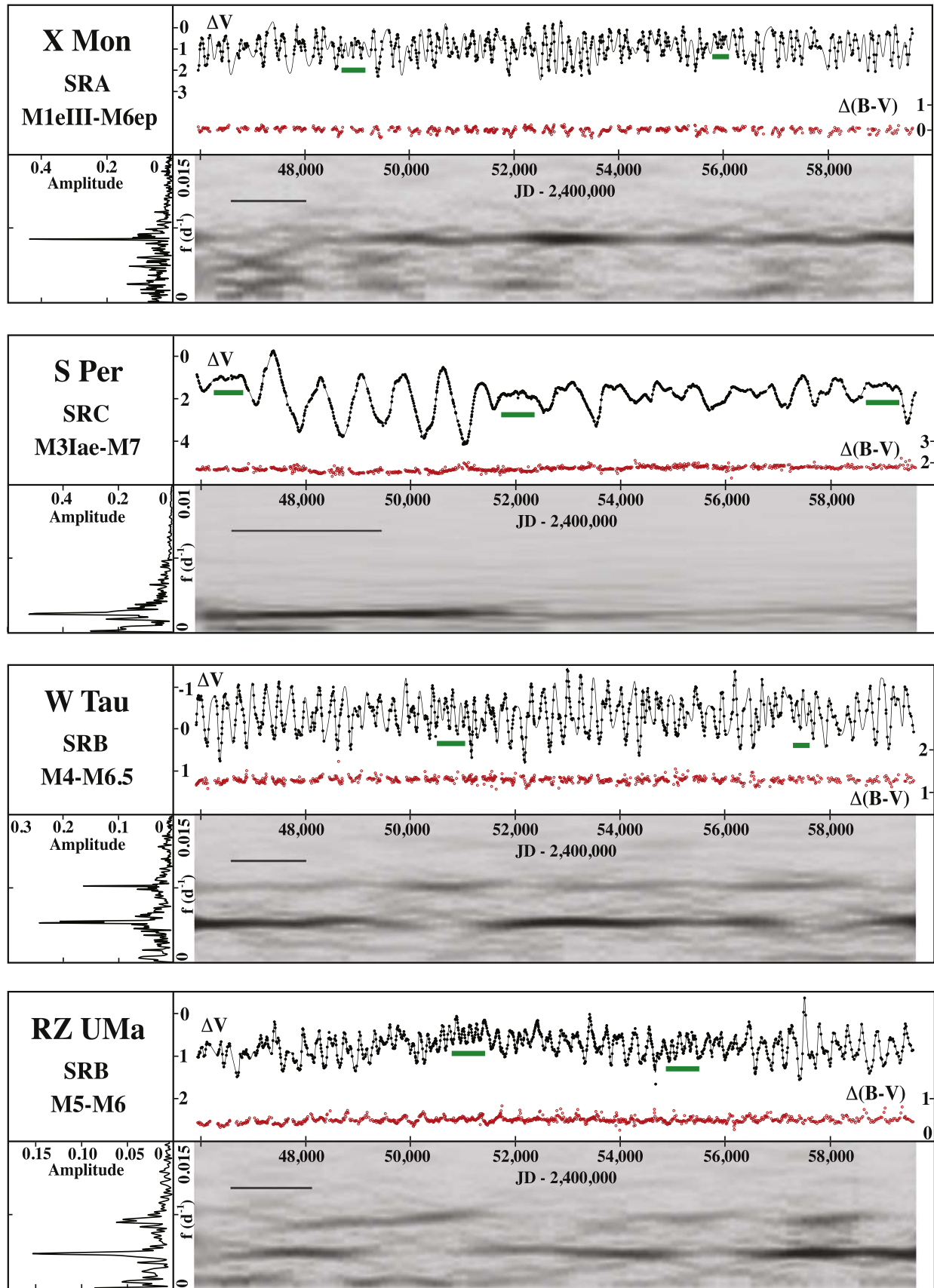


Figure 6. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

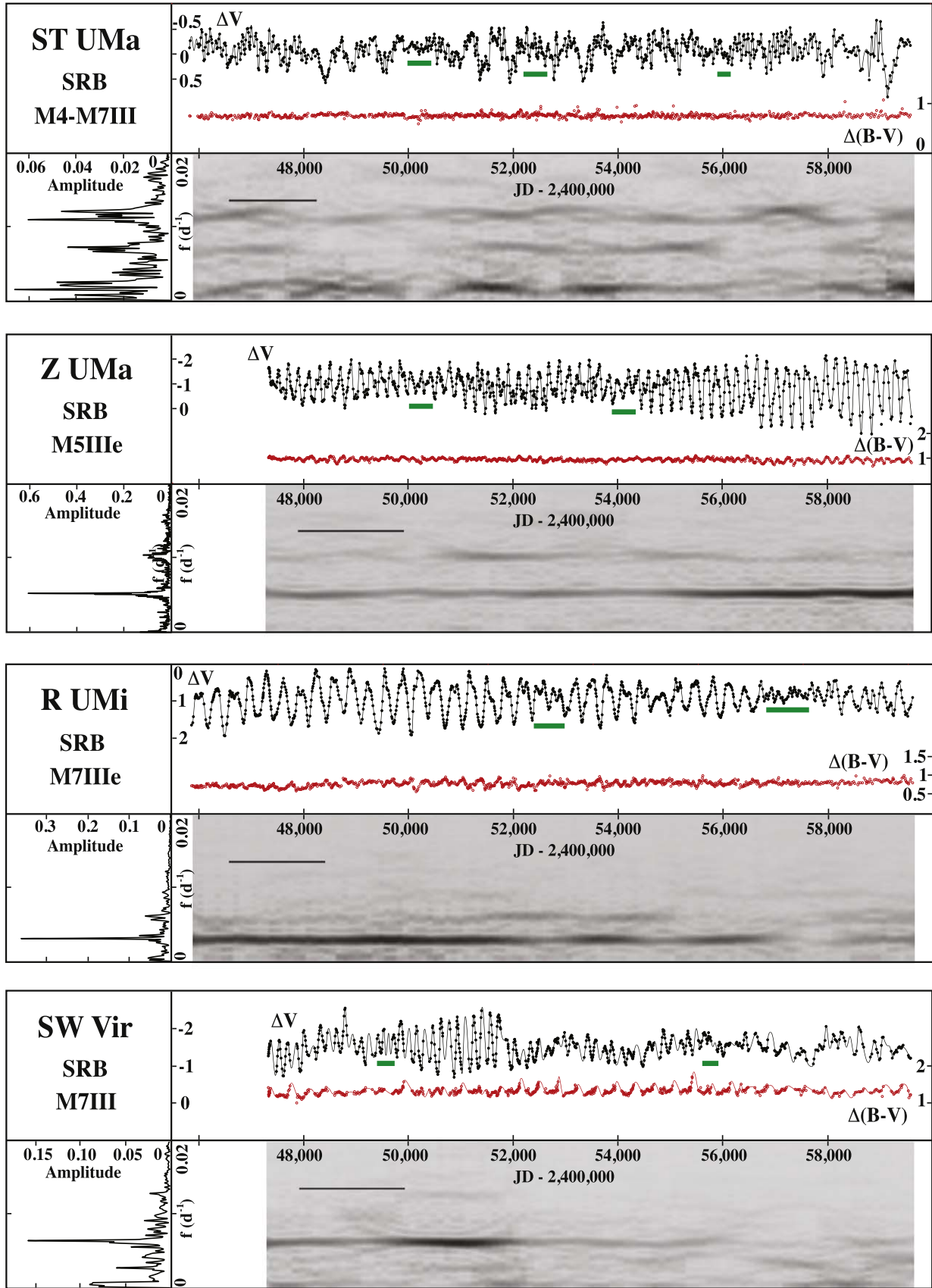


Figure 7. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are associated with a change in frequency. See Figure 1 and the text for details.

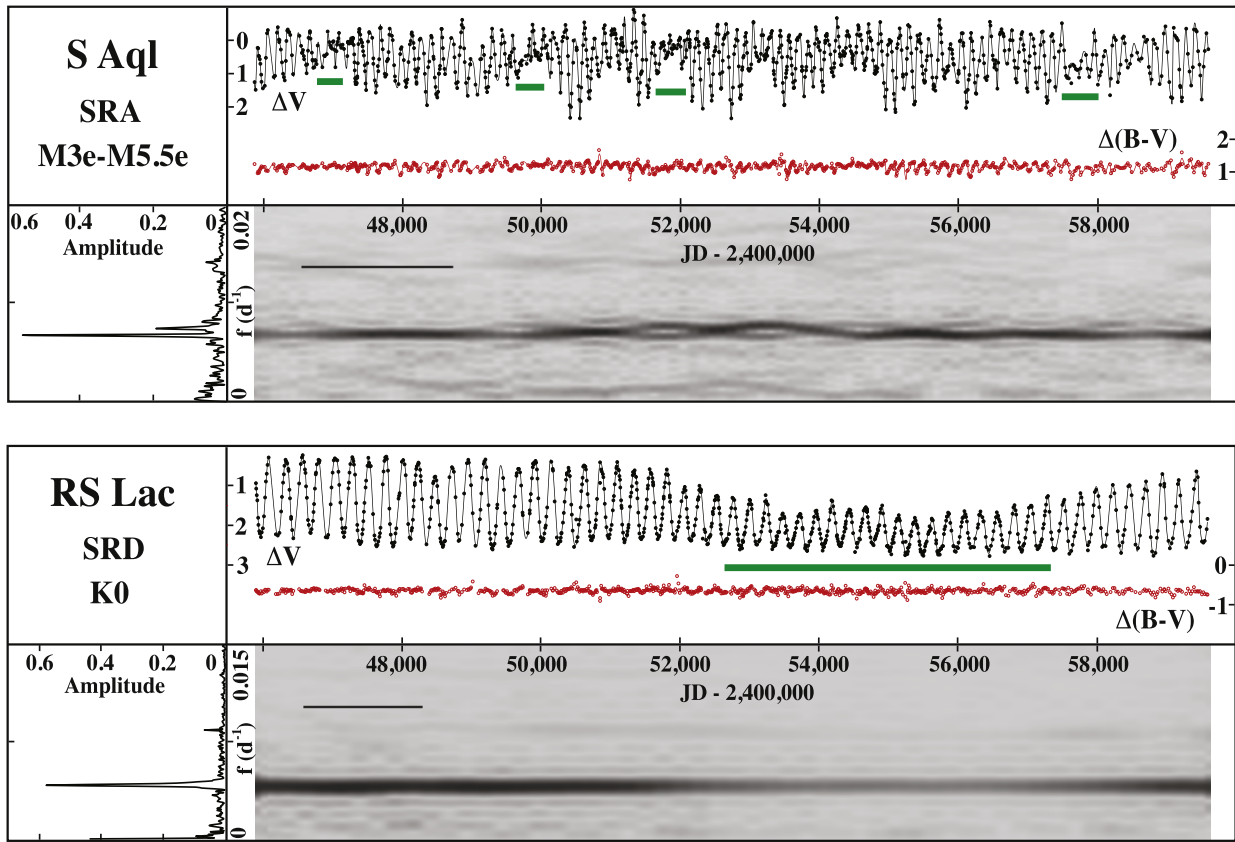


Figure 8. Photometric data and Fourier analysis results for stars with episodes of reduced amplitude that are not associated with a change in frequency. See Figure 1 and the text for details.

for which the data were less reliable and had less influence on the fit. This scheme was found to give a better overall representation of the data, which often have complex variations, than did the entirely computational fitting processes that were tried. A number of numerical experiments showed that various ways to fit over any small gaps in the data had very little effect on the Fourier analysis results. The results presented here are based on analyzing magnitude data, but tests with the corresponding intensity data gave very similar results.

The investigation of rapid, small-amplitude variations in the light curves requires photometric data with reasonable precision as well as good time resolution. This is difficult to accomplish with visual data because the binning that is required to reduce the scatter in the data also smooths over short-timescale variations. The advantage of photometric observations in this situation is illustrated in Figure 11, which compares a small sample of the Grinnell data for R UMi with the corresponding AAVSO data both with and without binning. The photometric ΔV data have been converted to V by adding the comparison star value of 8.47. The AAVSO data have been binned using the 7 day interval recommended by AAVSO for these stars² and 0.29 mag has been subtracted to adjust the visual data to V as described in Cadmus (2021). Although binned visual data can reveal the nature of subtle light-curve details in some cases, the photometric data clearly provide a much more solid foundation for exploring these variations.

These data will be archived with the AAVSO.

4. Fourier Analysis

Each of the ΔV light curves was Fourier analyzed in two ways: as a complete data set to produce a conventional Fourier spectrum and with a moving window to produce a Fourier map. The first approach gives more precise frequencies while the second reveals the time dependence of the Fourier components, but with the reduced frequency resolution imposed by the “uncertainty principle” and the limited duration of the data segments. The Fourier transforms were performed using software based on “slow_ft” (S. Kawaler 1996, private communication), which does a simple transform and takes advantage of the dense, equally spaced data produced by the spline fitting procedure. For the analysis of entire light curves, the frequency range and the number of frequency points in the spectrum were optimized for each star. The peak heights in the spectra indicate the amplitudes of each frequency component. These Fourier spectra are shown in the lower left corner of each of the panels in Figures 1–10. These plots are oriented and scaled to facilitate comparison with the 2D Fourier maps described below. As expected from the diversity and complexity of the light curves, there is a great deal of variation in the Fourier spectra. The frequency of the dominant low-frequency peak for each star during the interval of these observations, the corresponding period, and the frequency ratios for the two most significant higher-frequency peaks relative to the dominant low-frequency peak are presented in Table 1. The complex time variation in the frequencies for ST UMa (see below) blurs the peaks in the Fourier spectrum so the periods were derived from estimates of the centers of those clumps of peaks. In a few cases peaks in the Fourier spectra are double; these cases are

² <https://www.aavso.org/quickstart-guide-using-aavso-data>, Mean Curves.

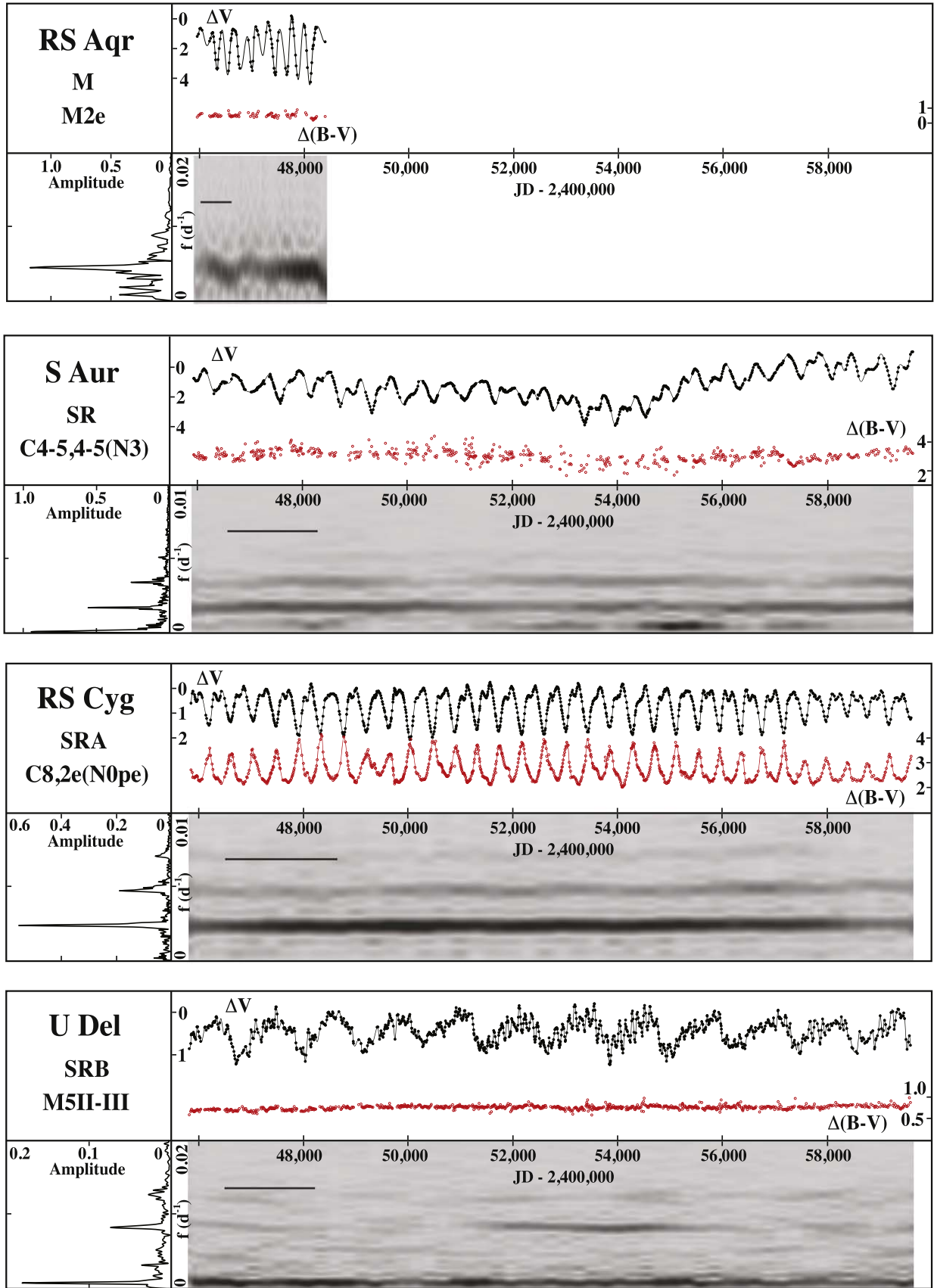


Figure 9. Photometric data and Fourier analysis results for stars that do not exhibit episodes of reduced amplitude. See Figure 1 and the text for details.

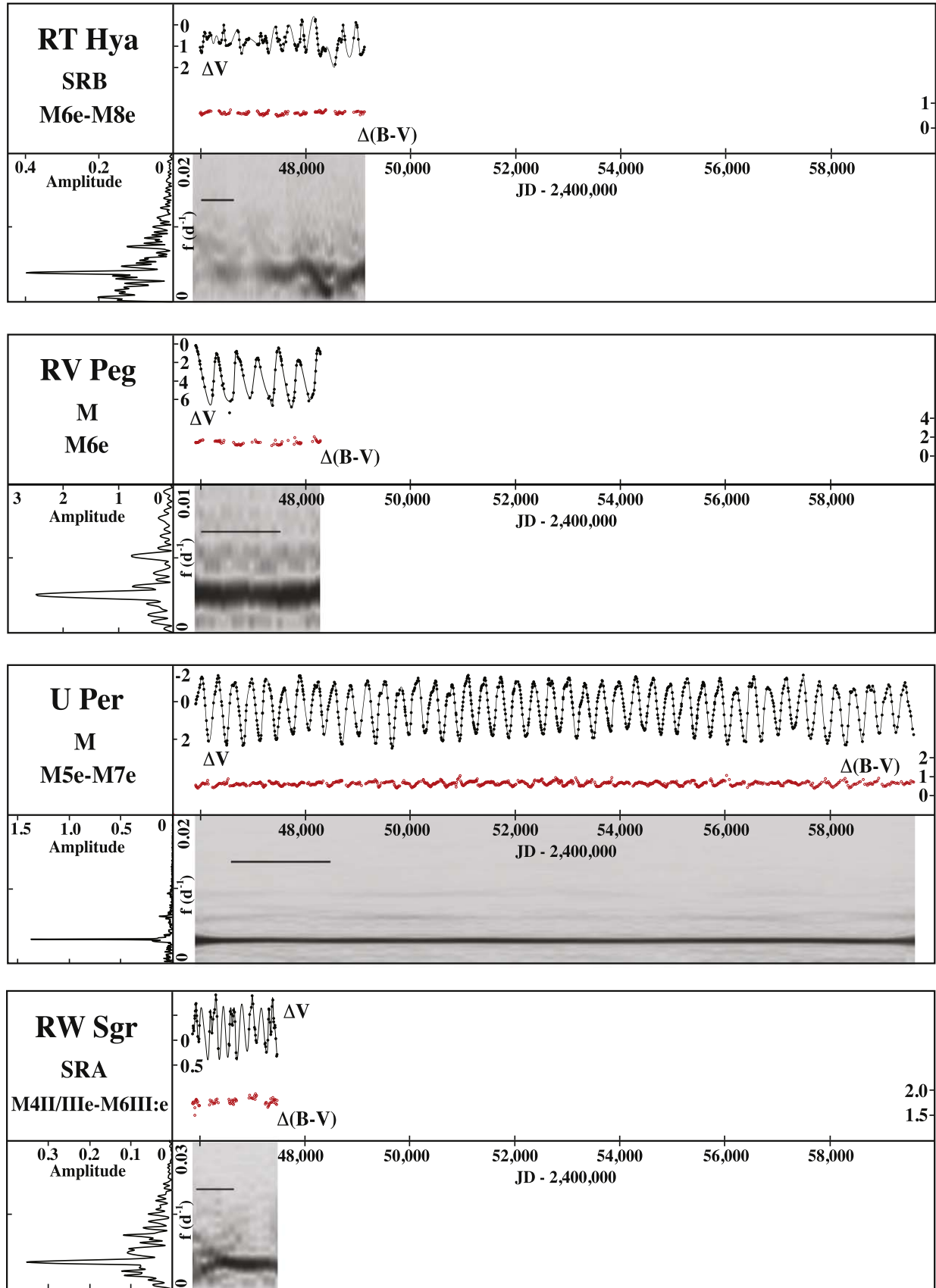


Figure 10. Photometric data and Fourier analysis results for stars that do not exhibit episodes of reduced amplitude. See Figure 1 and the text for details.

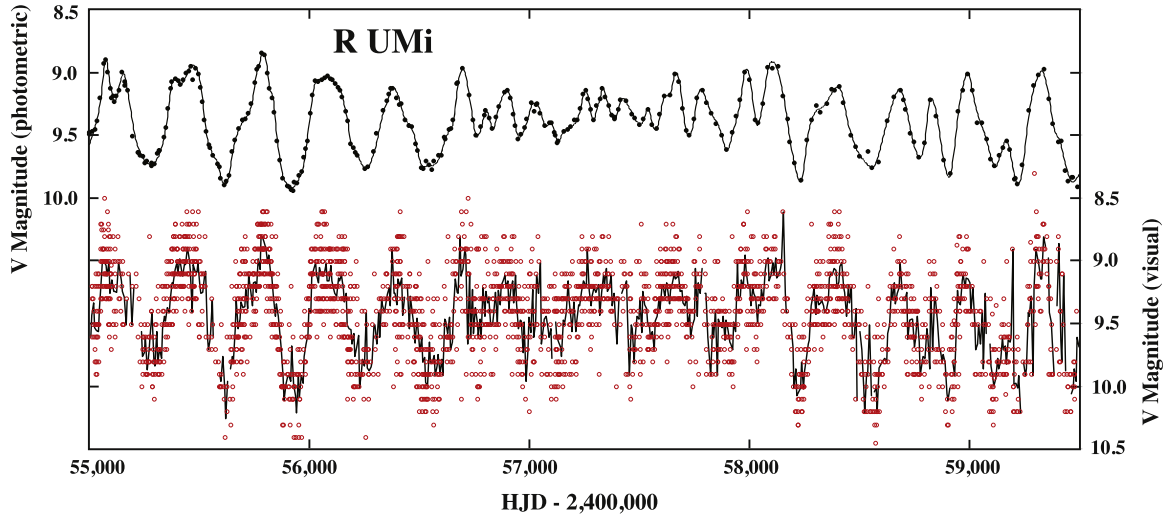


Figure 11. The Grinnell photometric ΔV data, converted to V , for a sample of the R UMi light curve (solid symbols), the spline fit to the photometric data (line), the raw AAVSO visual data adjusted to V as described in the text (open red symbols), and the AAVSO data binned by 7 days (line).

indicated in Table 1 and the values shown are averages of the two values in the spectra.

The Fourier maps (ΔV amplitude versus both frequency and time) provide a way to visualize how the strengths of the contributions of various frequencies of the light-curve change over time. Those results are shown below the light curves in Figures 1–10 using the same JD scale to facilitate comparison between the light curves and the maps. The frequency scales of the Fourier spectra of the entire light curves described earlier are also aligned with the maps to make the relationships clear. These maps were generated using the Fourier analysis scheme described above, but applied to the data in many time windows of limited duration that were progressively offset to span the full range of the data. Note this is not the same as a wavelet transform although the objectives are similar. The windows typically overlapped by about 10% or 20% of their widths and were Gaussian in shape to reduce artifacts that arise from a rectangular window shape. The width of the Gaussian window and the overlap between windows were adjusted for each star to achieve an optimum trade-off of frequency and time resolution. The bar in the upper left corner of each map is the FWHM of the Gaussian window. Maps made with narrower windows have worse frequency resolution but show that the transitions are a bit more abrupt than is apparent in the optimized maps. Both types of Fourier analysis presented here were performed on magnitude data but the results obtained by analyzing intensity data were not dramatically different.

Fourier analysis was also used to investigate the possible contribution of stochastic processes to the light curves by fitting Lorentzian curves to the clumps of peaks that are often present in the Fourier spectra. This was done with Fourier spectra generated with higher resolution than those presented in Figures 1–10. The half-width at half-maximum (HWHM) values of these Lorentzians were used to calculate lifetimes for the modes as $(2\pi\text{HWHM})^{-1}$ (Toutain & Fröhlich 1992; Bedding et al. 2005).

5. Results

The ΔV light curves in Figures 1–10 show a variety of types of variation in addition to the basic pulsation. These include

slow variations in amplitude, relatively rapid changes in amplitude, drifts in mean brightness, and generally erratic behavior. In most cases there is little variation in $\Delta(B - V)$ but for the carbon stars (EU And, U Cam, WZ Cas, RS Cyg, TT Cyg, V778 Cyg, RY Dra, and UX Dra) the $\Delta(B - V)$ variation mimics the ΔV variation, indicating that the magnitude change in B is twice that in V . S Aur is also a carbon star but it had a B magnitude around 16 so the $\Delta(B - V)$ data are marginal and the details are less apparent. The carbon stars in this study have amplitudes that generally increase with period, a trend that is not evident for the other stars.

The Fourier spectra of the full light curves and the Fourier maps reflect the complexity of the light curves and show that the frequency compositions of those light curves are often highly variable. The complexity of the light curves of these stars means that the astrophysically interesting pulsational period may differ from the “observational” period that characterizes the overall light variation of the star and is presumably the period given in the AAVSO International Variable Star Index database.³ The emphasis of this paper is on modes of pulsation so the periods shown in Table 1 are those that seem most likely to be relevant in that context, although they may not always be likely candidates for the fundamental period. The long secondary periods that often appear as very low-frequency peaks were not selected even if they dominate the light variation. The long secondary periods that appear as slow variations in the light curves of many long period variable stars have been the subject of much study and debate. Wood et al. (2004) reviewed various explanations and concluded that none seemed viable, but more recently Soszyński (2022) summarized the state of work on this problem and provided evidence that the cause is a brown dwarf and associated dusty cloud in orbit around the red giant. The variety and complexity of the light curves makes classification of them an approximate process, but in most cases there is a clear, usually dominant, lowest-frequency peak that corresponds to the star’s apparent period of variation and is in good agreement with the AAVSO period, and additional peaks appear to be harmonics of that. In several cases (WZ Cas, RU Cyg, TT Cyg, W Cyg, UX Dra,

³ <http://www.aavso.org/vsx>

X Lib, and RZ UMa) the period reported here is approximately twice the AAVSO period, suggesting that the AAVSO period may be a harmonic although that relationship is not always as tight as in the previous cases. For a few stars (RX Boo, U Del, X Mon, and S Per) there is good agreement with the AAVSO period but the other peaks are not harmonic or there is only one significant peak. For U Cam and RY Dra the periods reported here are significantly different from the AAVSO values and other peaks do not appear to be harmonics. RX Boo, U Cam, V CVn, W Cyg, U Del, RS Lac, and SW Vir, show evidence of long secondary periods. S Aur has a slow variation that is probably not a conventional long secondary period.

Several stars have particularly interesting behavior. The S Aql ΔV light curve shows a nearly periodic variation in amplitude with a period of about 1100 days and the Fourier spectrum shows partial splitting of the dominant frequency component that corresponds to a period of about 1380 days. While these values are approximately consistent with this amplitude variation being a beat phenomenon, it is hard to distinguish two physically separate processes at similar frequencies from a process that directly modulates the amplitude, resulting in a doublet in the Fourier spectrum. The 410 day period of the dominant peak in the spectrum of RW Boo, which corresponds to the obvious large oscillations, is approximately twice that of the small 198 day peak, which is close to the AAVSO period but does not correspond to an obvious oscillation in our light curve. The nature of the 57 day peak is unclear. The large peak at 161 days in the RX Boo spectrum is close to the AAVSO value. This frequency and that of the small peak corresponding to a period of 99 days are nearly harmonics of a 307 day period for which there is only a vague hint of a peak. For U Cam AAVSO gives a very long period (2800 days) that corresponds to the lowest large peak in our Fourier spectrum and is the long secondary period identified by Percy & Huang (2015). The large peak at 218 days may be a harmonic of the doublet peak near 394 days. The 120 day period in the spectrum of U Del is close to the AAVSO value and corresponds to the small, rapid oscillations, not the obvious large, slow oscillation that appears to be a long secondary period. This frequency and that of the 79 day peak are very close to harmonics of 238 days, although no such peak is apparent in the spectrum. X Her is an unusual case because the frequencies of the three strong peaks are approximately equally spaced but the higher frequencies are not integer multiples of the lowest strong frequency (not harmonic). The 102 day peak corresponds to the AAVSO period. The lowest-frequency (691 days) peak does not seem to be low enough to be a long secondary period. The ST UMa spectrum is messy and serves as a reminder that, observationally if not physically, stars with complex light curves do not have “a period.” The periods of the groups of peaks near 140, 86 days (similar to AAVSO), and 59 days are close to harmonics of 256 days, which appears only vaguely in the Fourier spectrum. The period of the 526 day group of peaks is very close to double that of the elusive 256 day group and corresponds to the obvious variation. A plot of these frequencies versus sequential integers forms a smooth, but not linear, upward trend. Assigning the peaks to the integers 1, 2, 4, 6, and 9 gives a nearly linear relationship but it seems unlikely that harmonics this high are involved.

There is a tendency for the frequency ratios to be harmonic, as shown in Table 1 and Figure 12, with the ratios of higher-

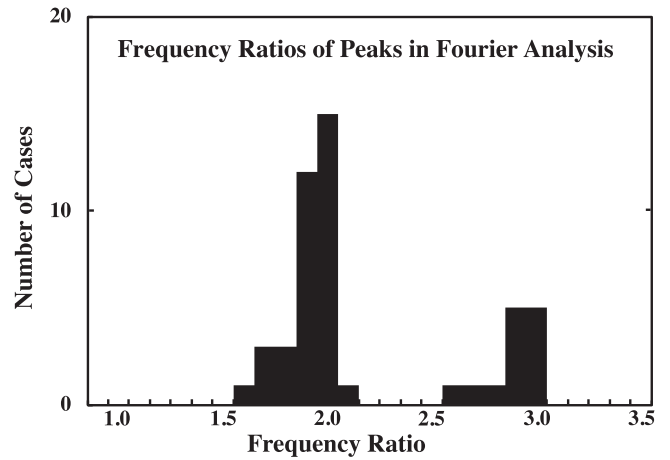


Figure 12. The ratios of the frequencies of the strongest peaks in the Fourier spectra to the lowest strong frequency. There is a tendency for the values to clump near integers.

frequency peaks to the lowest-frequency strong peak localized near 2 and 3. The lowest strong frequency is the fundamental in the context of Fourier analysis and probably corresponds to the fundamental mode of pulsation of the star. The interpretation of the other peaks will be discussed below.

The ΔV light curves can be divided into several approximate categories although some stars fall into more than one of them. In some cases the ΔV light curves show a significant degree of cycle-to-cycle consistency (although these stars are still only semiregular) and the Fourier spectra are relatively simple. The corresponding Fourier maps show the multiple frequencies but little time variation, as expected, and the higher frequencies never appear in the absence of the dominant lowest frequency. In other cases the light curves have substantial cycle-to-cycle variations and the Fourier spectra and maps show a great deal of structure that is dominated by amplitude variations. For most of these stars the variations that are apparent in the maps are not frequency changes, but situations in which the amplitude of a larger-amplitude component decreases, revealing a smaller-amplitude component that was present all along. Only RZ UMa shows any significant long-term drift in any of the frequencies.

In most cases the episodes of markedly reduced amplitude indicated by the green bars in the light-curve figures are accompanied by a change in the dominant frequency (Figures 1–7) or simply by a decrease in the amplitude of the dominant frequency (Figure 8). The ratios of the frequencies within and near these episodes show essentially the same concentration near integer values as do the frequency ratios for the light curve as a whole.

Often the dominant features in the Fourier spectra are not simple peaks, but are clumps of peaks that result from the temporal variations in the strengths of the frequency components. Sometimes the envelope of these clumps of peaks has an approximately Lorentzian shape, but in many other cases these clumps are flat-topped, dominated by a single peak, or have other characteristics that make representation by a Lorentzian unconvincing. Nevertheless, Lorentzians were fit to most of them and the widths were converted to mode lifetimes as described above, based on the assumption that these clumps result from stochastic excitation in the presence of damping. Except for a few cases the overall characteristics of the set are probably more meaningful than the values for individual stars. The lifetimes ranged from a couple of years to just over 10 yr

and the mean for the 26 cases (often more than one per star) that seemed most convincing was about 3 yr. There was no strong correlation with period but the stars with the most complex behavior in the Fourier maps tended to have shorter lifetimes. When more than one clump of peaks was present for a star the lifetimes were usually significantly different, but the results for higher-frequency clumps are often less reliable because the signal was usually weaker. V Boo was the case for which the Lorentzian fitting was the most convincing, with lifetimes for the two clumps of peaks of 2.8 and 1.8 yr. The two highest-frequency clumps for X Her were also reasonably Lorentzian but had very different lifetimes (0.7 and 1.8 yr).

6. Discussion

The most obvious features of our data—the wide variety of types of light-curve variations and the often complex time dependence of their frequency compositions—is not surprising. Of particular interest is what these variations can tell us about the role of damped stochastic excitation. As described in the Section 5, our investigation into the possibility of stochastic excitation in the stars in this study revealed some generic information but in only two cases, V Boo and X Her, do the results seem to warrant more detailed discussion. The fact that the clumps of peaks for some of these stars appear to be Lorentzian suggests that stochastic processes may be important for some of these stars but less important for others. Cunha et al. (2020) have found that the transition between significant stochastic excitation and primarily coherent excitation occurs at a period in the neighborhood of 60 days. The work of Yu et al. (2020) suggests a transition around 70 days. The periods of all of our stars are longer than this. The shortest is 86 days (ST UMa), the mean is 274 days, V Boo is at 260 days, and X Her is at 175 days. It is therefore not surprising that we do not see clear evidence for stochastic excitation in many of these stars. However, Bedding et al. (2005) have reported evidence for stochastic excitation in L₂ Pup, which has a period of about 140 days although Cunha et al. (2020) conclude that it is a “classical pulsator.” Another consideration is the fact that there is no obvious decrease in the complexity of the Fourier maps as the period increases as one might expect if that complexity, which leads to broadening of the peaks in the overall Fourier spectrum regardless of its cause, is the result of a completely stochastic process. Overall, our results are consistent with previous work that shows that stochastic excitation can be significant in some semiregular variable stars and that the mode lifetimes are typically a few years (Bedding 2003; Bedding et al. 2005; Kiss et al. 2006). It is possible that stochastic excitation is more important for higher-frequency components of the Fourier spectra that may correspond to higher pulsational modes but the data presented here are not sufficient to establish that. Of course, these are complicated stars and processes beyond those discussed here might be involved in the identification of the origin of the frequency components of the light curves.

The complex shapes of the light curves are the result of multiple frequency components that have been attributed to multiple modes of pulsation. However, it is sometimes unclear to what extent these components represent different modes of pulsation of the star and to what extent are they simply harmonics associated with the shape of the light curve. The frequency ratio near 2 is similar to what one might expect for either a harmonic or two modes (Ostlie & Cox 1986), although the frequency ratio near 3 seems less likely to be that between

two adjacent modes. Our observations that these ratios are, on average, about 97% of the integer values is consistent with earlier results (Mattei et al. 1997; Kiss et al. 1999; Percy & Tan 2013).

The Fourier maps show a continuum of degrees of complexity, defined as the extent to which there are abrupt changes in the frequency components, rather than several discrete types of behavior. The variation of complexity with variability type SRA–SRD is scattered. There might be a slight correlation with luminosity class, but our data are not sufficient to confirm that. The most interesting result is the relationship between complexity and spectral type or mean $B - V$. Most of the stars are similar in spectral type and populate the entire range of complexity. The carbon stars, on the other hand, have Fourier maps that are generally, but not universally, relatively complex. We are engaged in a program of long-term spectroscopy to investigate this phenomenon.

7. Conclusions

This work represents an unusually comprehensive investigation of the frequency composition of the light variations of semiregular variable stars, primarily because of the duration and precision of the data set. Several conclusions can be drawn from the inspection of our light curves and Fourier analysis results: (1) the light curves display a variety of kinds of variation that are different for different stars, (2) the variation in $\Delta(B - V)$ is generally small except in the case of the carbon stars, for which it is comparable to the variation in ΔV , (3) the light curves usually have multiple frequency components that tend to be in integer ratios, (4) the time dependence of the strengths of the frequency components is frequently very complex, including instances of sudden changes in the dominant frequency or changes in the strength of the dominant frequency, (5) the carbon stars in this sample seem to be more susceptible to erratic frequency behavior than are the other stars, and (6) there is evidence for stochastic excitation in some cases.

The common occurrence of near-integer frequency ratios could result from either multiple modes or harmonics of modes and there is evidence supporting each conclusion. This suggests that both processes are at work but it is not easy to separate them. The abrupt changes as well as more gradual variations in the frequency compositions may reflect the presence of damped stochastic excitation.

Acknowledgments

It is a great pleasure to thank many Grinnell College undergraduate students for their assistance with this project and Lee Anne Willson for her valuable insight. Valuable comments by a referee are also gratefully acknowledged. Financial support was provided by the National Science Foundation, Research Corporation, the American Astronomical Society, Sigma Xi, and Grinnell College. This research has made use of the SIMBAD database, operated at CDS, Strasbourg, France.

ORCID iDs

Robert R. Cadmus, Jr.  <https://orcid.org/0000-0003-1921-7421>

References

- Bányai, E., Kiss, L. L., Bedding, T. R., et al. 2013, *MNRAS*, **436**, 1576
- Bedding, T. R. 2003, *Ap&SS*, **284**, 61
- Bedding, T. R., Kiss, L. L., Kjeldsen, H., et al. 2005, *MNRAS*, **361**, 1375
- Bedding, T. R., & Kjeldsen, H. 2003, *PASA*, **20**, 203
- Cadmus, R. R., Jr 2021, *AJ*, **161**, 75

- Cadmus, R. R. 2022, [AJ](#), **163**, 265
- Christensen-Dalsgaard, J., Kjeldsen, H., & Mattei, J. A. 2001, [ApJL](#), **562**, L141
- Cunha, M. S., Avelino, P. P., & Chaplin, W. J. 2020, [MNRAS](#), **499**, 4687
- Dziembowski, W. A., Gough, D. O., Houdek, G., & Sienkiewicz, R. 2001, [MNRAS](#), **328**, 601
- Høg, E., Fabricius, C., Makarov, V. V., et al. 2000, [A&A](#), **355**, L27
- Kiss, L. L., Szabó, G. M., & Bedding, T. R. 2006, [MNRAS](#), **372**, 1721
- Kiss, L. L., Szatmáry, K., Cadmus, R. R., Jr., & Mattei, J. A. 1999, [A&A](#), **346**, 542
- Kjeldsen, H., & Bedding, T. R. 1995, [A&A](#), **293**, 87
- Mattei, J. A., & Foster, G. 1997, [BAAS](#), **29**, 1284
- Mattei, J. A., Foster, G., Hurwitz, L. A., et al. 1997, in ESA SP-402, *Hipparcos — Venice '97*, ed. R. M. Bonnet et al. (Noordwijk: ESA), 269
- Mosser, B., Dziembowski, W. A., Belkacem, K., et al. 2013, [A&A](#), **559**, A137
- Ostlie, D. A., & Cox, A. N. 1986, [ApJ](#), **311**, 864
- Percy, J. R., Besla, G., Velocci, V., & Henry, G. W. 2003, [PASP](#), **115**, 479
- Percy, J. R., & Huang, D. J. 2015, [JAAVSO](#), **43**, 118
- Percy, J. R., & Polano, S. 1998, in *ASP Conf. Ser. 135, A Half Century of Stellar Pulsation Interpretation*, ed. P. A. Bradley & J. A. Guzik (San Francisco, CA: ASP), 249
- Soszyński, I. 2022, in *XL Polish Astronomical Society Meeting 12*, ed. E. Szuszkiewicz et al. (Warsaw: Polish Astronomical Society), 154
- Percy, J. R., & Tan, P. J. 2013, [JAAVSO](#), **41**, 1
- Toutain, T., & Fröhlich, C. 1992, [A&A](#), **257**, 287
- Wood, P. R., Olivier, E. A., & Kawaler, S. D. 2004, [ApJ](#), **604**, 800
- Yu, J., Bedding, T. R., Stello, D., et al. 2020, [MNRAS](#), **493**, 1388